

OFFICE NETWORK BUILD

Default Routing

After successfully implementing static routing, the company's different networks can now communicate with one another through manually configured routes. However, as the network continues to grow, manually creating a specific route for every possible destination can become inefficient and difficult to maintain.

In the current network, the routers already use a transit network to communicate with each other. We can build upon this existing topology by introducing default routing, which provides a route for destinations that are not explicitly listed in the routing table.

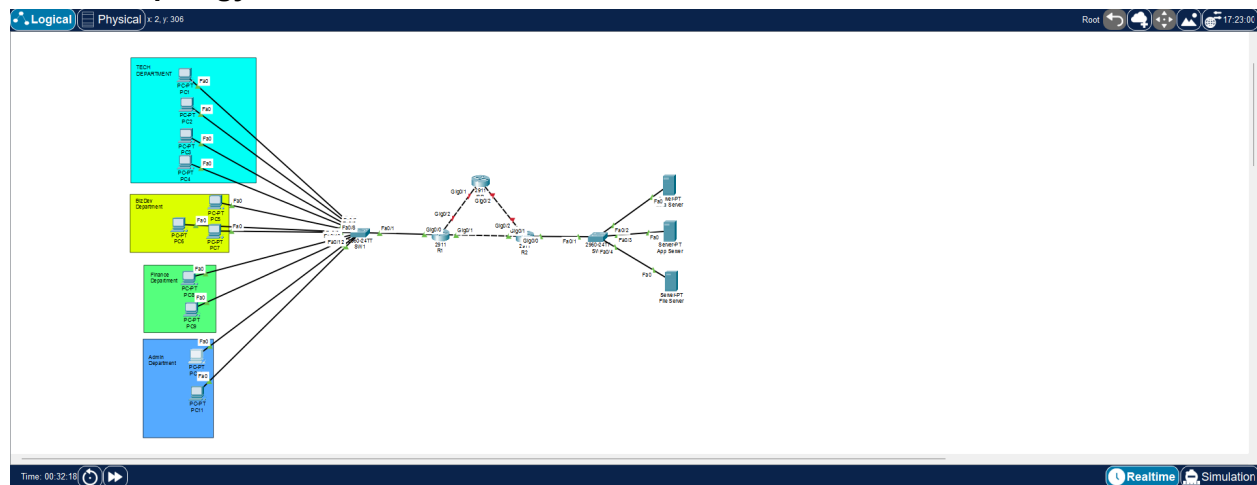
For this project, default routing will represent the next stage of the company's network expansion. Instead of configuring individual routes for every unknown or external network, the router can forward such traffic toward an upstream router through the existing transit link.

This demonstrates an important principle in real-world networking: routers do not always need to know every destination individually; sometimes, they only need to know where to send traffic they don't have a specific route for.

The image below shows the current IP configurations including VLAN assignments and interfaces used.

Office Network Build					
Device	IP	Subnet	Default Gateway	VLAN	Interface
R1	192.168.100.1	255.255.255.248 - /29	N/A	VLAN 10	G0/0.10
R1	192.168.100.9	255.255.255.248 - /29	N/A	VLAN 20	G0/0.20
R1	192.168.100.17	255.255.255.248 - /29	N/A	VLAN 30	G0/0.30
R1	192.168.100.25	255.255.255.248 - /29	N/A	VLAN 40	G0/0.40
R1 (Transit Link)	192.168.102.1	255.255.255.252 - /30	N/A	N/A	G0/1
R2	192.168.101.1	255.255.255.0 - /24	N/A	N/A	G0/0
R2 (Transit Link)	192.168.102.2	255.255.255.252 - /30	N/A	N/A	G0/1
PC1	192.168.100.2	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC2	192.168.100.3	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC3	192.168.100.4	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC4	192.168.100.5	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC5	192.168.100.10	255.255.255.248 - /29	192.168.100.9	VLAN 20	Fa0
PC6	192.168.100.11	255.255.255.248 - /29	192.168.100.9	VLAN 20	Fa0
PC7	192.168.100.12	255.255.255.248 - /29	192.168.100.9	VLAN 20	Fa0
PC8	192.168.100.18	255.255.255.248 - /29	192.168.100.17	VLAN 30	Fa0
PC9	192.168.100.19	255.255.255.248 - /29	192.168.100.17	VLAN 30	Fa0
PC10	192.168.100.26	255.255.255.248 - /29	192.168.100.25	VLAN 40	Fa0
PC11	192.168.100.27	255.255.255.248 - /29	192.168.100.25	VLAN 40	Fa0
Data Server	192.168.101.10	255.255.255.0 - /24	192.168.101.1	N/A	Fa0
Application Server	192.168.101.20	255.255.255.0 - /24	192.168.101.1	N/A	Fa0
File Server	192.168.101.30	255.255.255.0 - /24	192.168.101.1	N/A	Fa0

Current Topology:



R1 Routing Table:

```

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

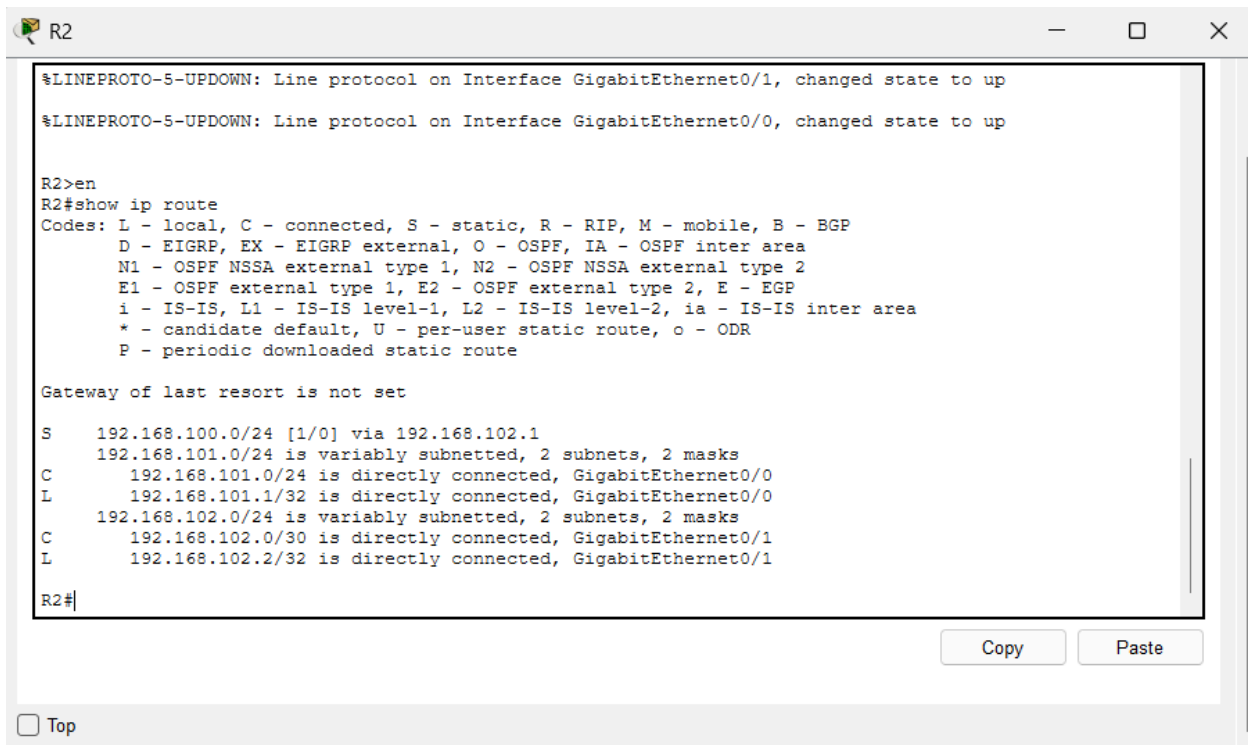
  192.168.100.0/24 is variably subnetted, 8 subnets, 2 masks
C       192.168.100.0/29 is directly connected, GigabitEthernet0/0.10
L       192.168.100.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.100.8/29 is directly connected, GigabitEthernet0/0.20
L       192.168.100.9/32 is directly connected, GigabitEthernet0/0.20
C       192.168.100.16/29 is directly connected, GigabitEthernet0/0.30
L       192.168.100.17/32 is directly connected, GigabitEthernet0/0.30
C       192.168.100.24/29 is directly connected, GigabitEthernet0/0.40
L       192.168.100.25/32 is directly connected, GigabitEthernet0/0.40
S       192.168.101.0/24 [1/0] via 192.168.102.2
  192.168.102.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.102.0/30 is directly connected, GigabitEthernet0/1
L       192.168.102.1/32 is directly connected, GigabitEthernet0/1

R1#
  
```

Copy Paste

☐ Top

R2 Routing Table:



```
R2
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2>en
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    192.168.100.0/24 [1/0] via 192.168.102.1
    192.168.101.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.101.0/24 is directly connected, GigabitEthernet0/0
L    192.168.101.1/32 is directly connected, GigabitEthernet0/0
    192.168.102.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.102.0/30 is directly connected, GigabitEthernet0/1
L    192.168.102.2/32 is directly connected, GigabitEthernet0/1

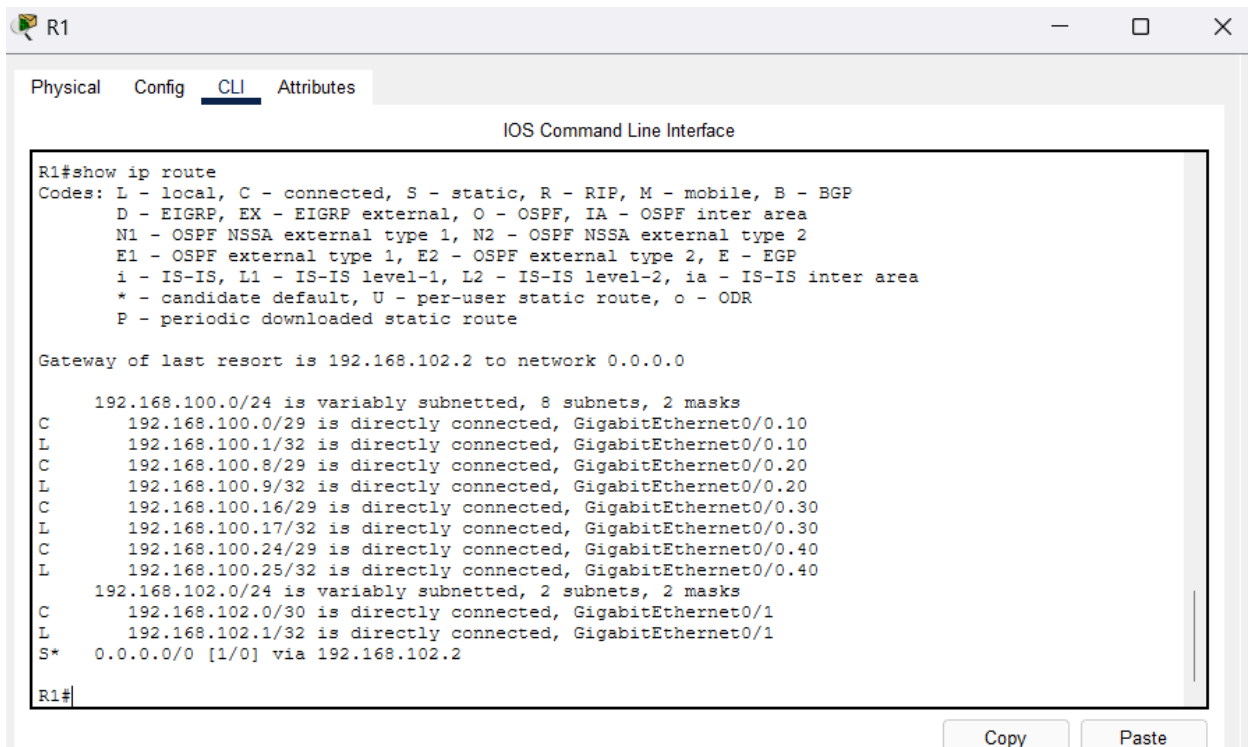
R2#
```

Copy Paste

☐ Top

Default routes are needed to make the network more efficient and easier to manage as it grows. While static routes work well when there are only a few known destination networks, they require administrators to manually configure a separate route for each destination. This can become difficult to maintain when the network expands or when there are many external networks that the router does not need to know individually. A default route provides a single route of last resort for all destinations that do not have a more specific entry in the routing table. Instead of maintaining numerous static routes, the router can simply forward unknown traffic to an upstream router or gateway. This reduces configuration complexity, makes the routing table easier to manage, and provides a more scalable approach for networks with a clear path toward external or unknown destinations.

R1 Routing Table (Static Route removed):



The screenshot shows the R1 CLI window with the 'CLI' tab selected. The command 'show ip route' has been executed, displaying the routing table. The output includes a legend for route codes, a gateway of last resort, and a list of routes. The routes are categorized by their source: C (connected), L (local), and S* (static). The static route is 0.0.0.0/0 [1/0] via 192.168.102.2.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.102.2 to network 0.0.0.0

 192.168.100.0/24 is variably subnetted, 8 subnets, 2 masks
C    192.168.100.0/29 is directly connected, GigabitEthernet0/0.10
L    192.168.100.1/32 is directly connected, GigabitEthernet0/0.10
C    192.168.100.8/29 is directly connected, GigabitEthernet0/0.20
L    192.168.100.9/32 is directly connected, GigabitEthernet0/0.20
C    192.168.100.16/29 is directly connected, GigabitEthernet0/0.30
L    192.168.100.17/32 is directly connected, GigabitEthernet0/0.30
C    192.168.100.24/29 is directly connected, GigabitEthernet0/0.40
L    192.168.100.25/32 is directly connected, GigabitEthernet0/0.40
 192.168.102.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.102.0/30 is directly connected, GigabitEthernet0/1
L    192.168.102.1/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 192.168.102.2

R1#
```

R2 Routing Table (Static Route removed):



The screenshot shows the R2 CLI window with the 'CLI' tab selected. The command 'show ip route' has been executed, displaying the routing table. The output includes a legend for route codes, a gateway of last resort, and a list of routes. The routes are categorized by their source: C (connected), L (local), and S* (static). The static route is 0.0.0.0/0 [1/0] via 192.168.102.1.

```
R2(config)#no ip route 192.168.100.0 255.255.255.0 192.168.102.1
R2(config)#ip route 0.0.0.0 0.0.0.0 192.168.102.1
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

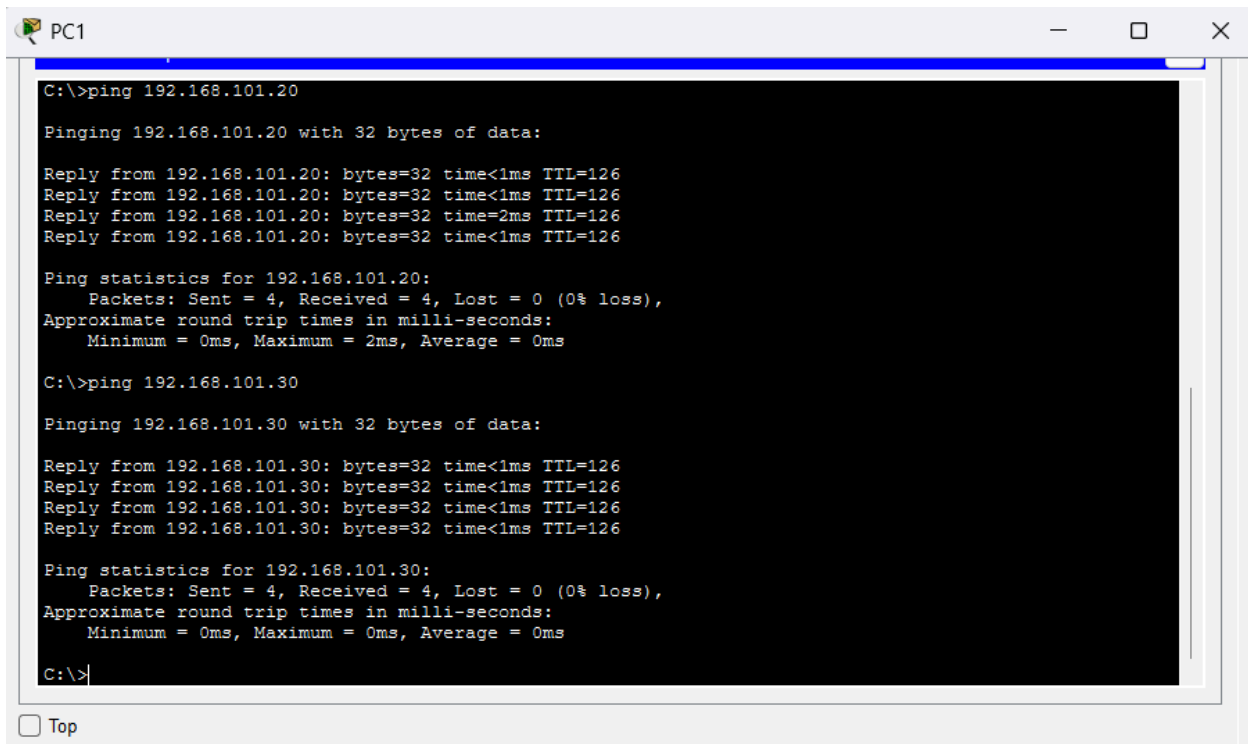
Gateway of last resort is 192.168.102.1 to network 0.0.0.0

 192.168.101.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.101.0/24 is directly connected, GigabitEthernet0/0
L    192.168.101.1/32 is directly connected, GigabitEthernet0/0
 192.168.102.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.102.0/30 is directly connected, GigabitEthernet0/1
L    192.168.102.2/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 192.168.102.1

R2#
```

We removed the previously configured static route to the server network to clearly demonstrate the purpose and behavior of default routing. Unlike the static route, which specifically identifies a particular destination network, the default route (0.0.0.0/0) acts as a route of last resort for any destination that does not have a more specific entry in the routing table. By removing the specific static route, R1 is now forced to rely on the default route when forwarding traffic to networks it does not directly know, allowing us to clearly observe how default routing simplifies routing configuration and provides a more efficient way of handling unknown or external destinations.

PC1 (Tech Dept.) ping to servers:



```
C:\>ping 192.168.101.20

Pinging 192.168.101.20 with 32 bytes of data:

Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time=2ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 192.168.101.30

Pinging 192.168.101.30 with 32 bytes of data:

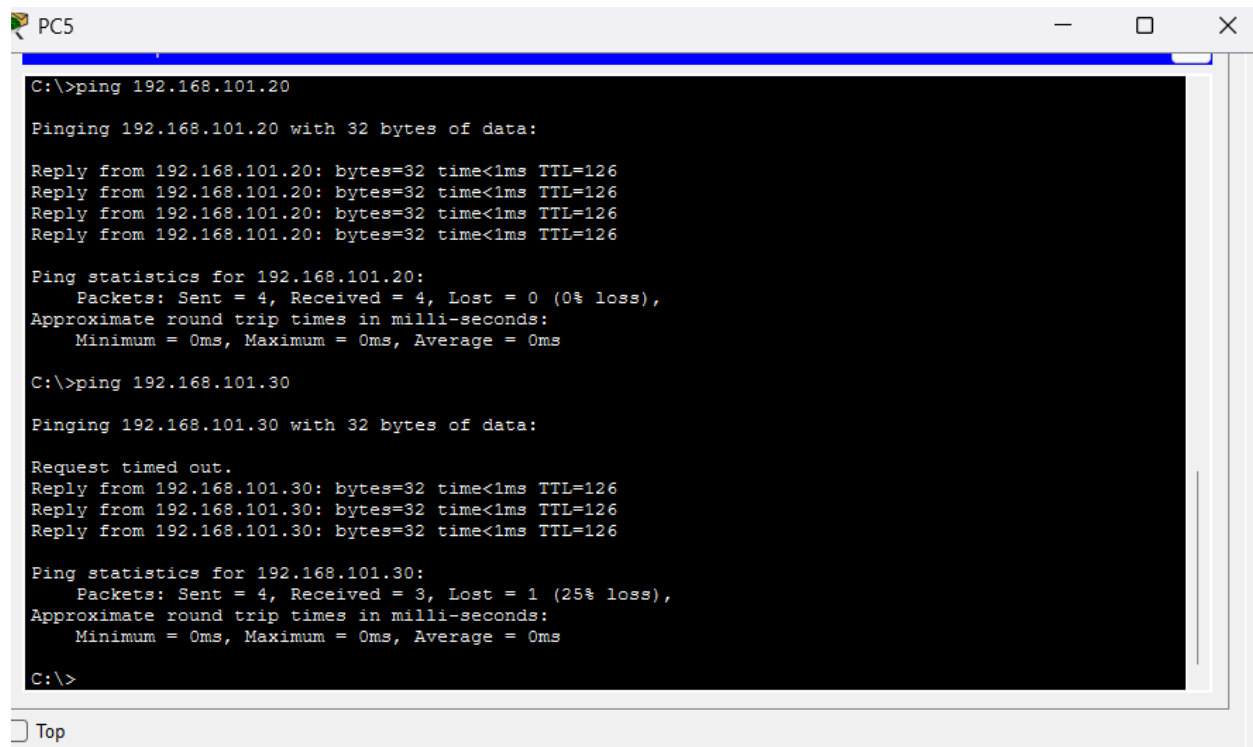
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

☐ Top

PC5 (BizDev Dept.) ping to servers:



```
C:\>ping 192.168.101.20

Pinging 192.168.101.20 with 32 bytes of data:

Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.101.30

Pinging 192.168.101.30 with 32 bytes of data:

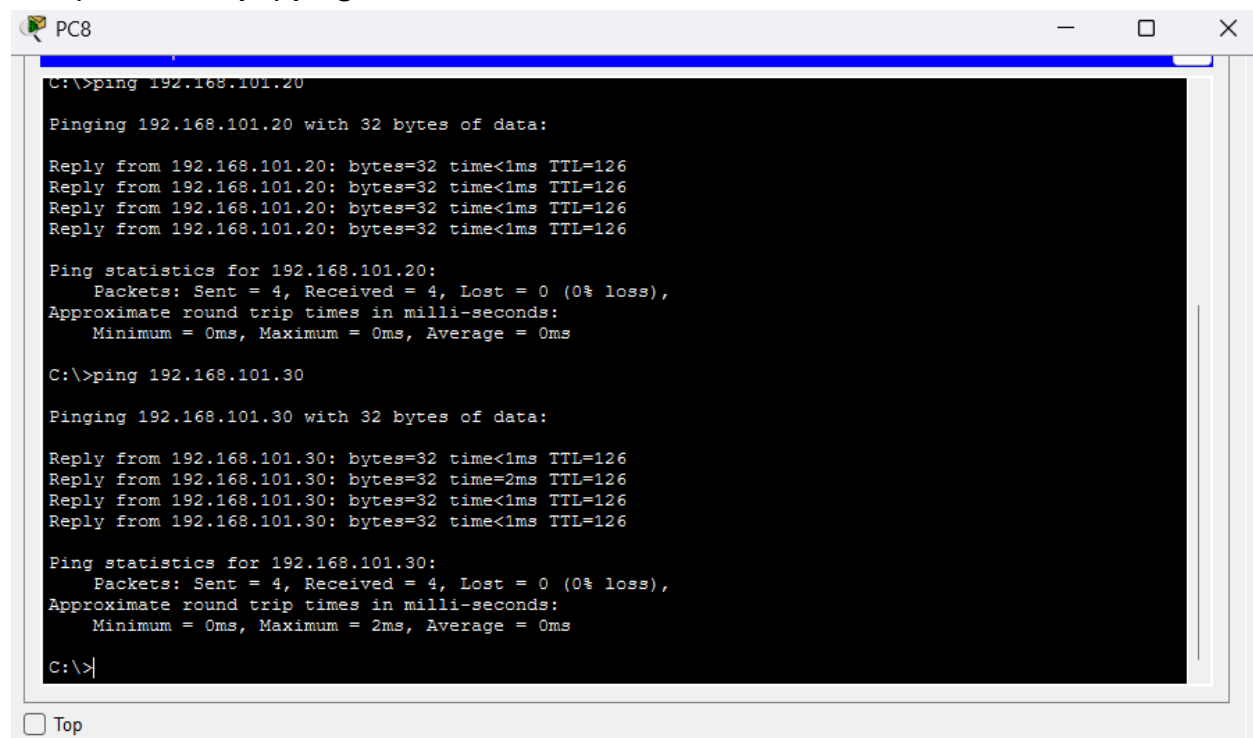
Request timed out.
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

☐ Top

PC8 (Finance Dept.) ping to servers:



```
C:\>ping 192.168.101.20

Pinging 192.168.101.20 with 32 bytes of data:

Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.101.30

Pinging 192.168.101.30 with 32 bytes of data:

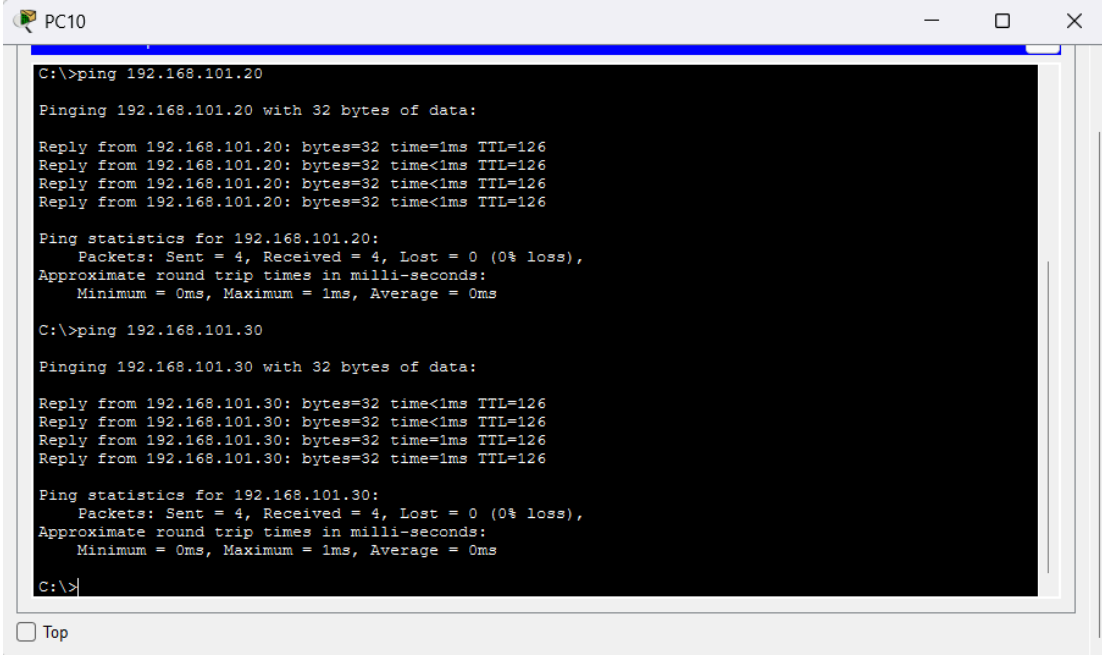
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time=2ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>|
```

☐ Top

PC10 (Admin Dept.) ping to servers:



```
C:\>ping 192.168.101.20

Pinging 192.168.101.20 with 32 bytes of data:

Reply from 192.168.101.20: bytes=32 time=1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.101.30

Pinging 192.168.101.30 with 32 bytes of data:

Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time=1ms TTL=126
Reply from 192.168.101.30: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.101.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Default Routes vs. Static Routes

While both static and default routes are manually configured by a network administrator, they serve different purposes. A static route provides a specific path to a particular destination network, making it useful when the network administrator needs precise control over how traffic reaches known networks. A default route, on the other hand, provides a general path for destinations that are not specifically listed in the routing table. In our project, the static route to the server network was removed and replaced with a default route to demonstrate this difference. The successful connectivity from the different departmental PCs to the servers showed that R1 could use the default route to forward traffic toward R2 when no more specific route was available.

We configured the routers again to have both Default and Static routes, is this redundancy?

After demonstrating default routing, we configured the routers again to have both specific static routes and a default route. However, having both routes does not automatically provide redundancy because they serve different purposes and the more specific static route will be preferred when its destination matches. In this configuration, the static route provides a specific path to a known network, while the default route serves as a route of last resort for destinations without a more specific entry. True routing redundancy would require an alternative path to the destination, such as a second router or link, so that traffic can still be forwarded if the primary path becomes unavailable.

Conclusion

The implementation of default routing builds upon the static routing configuration established earlier in the project. By removing the specific static route and configuring a default route, we demonstrated how a router can forward traffic toward an upstream router when no more specific destination exists in its routing table. The successful connectivity between the departmental PCs and the remote server network confirmed that the default route was functioning as intended. We then restored the static route alongside the default route to compare their roles and clarify that having both does not automatically provide redundancy. Overall, this exercise demonstrated how static and default routes can work together and provided a foundation for the next stage of the project: dynamic routing, where routers can learn network paths automatically.